

2024

State of Developer Ecosystem

2024

Language Trends, Satisfaction, and Strategic Shifts

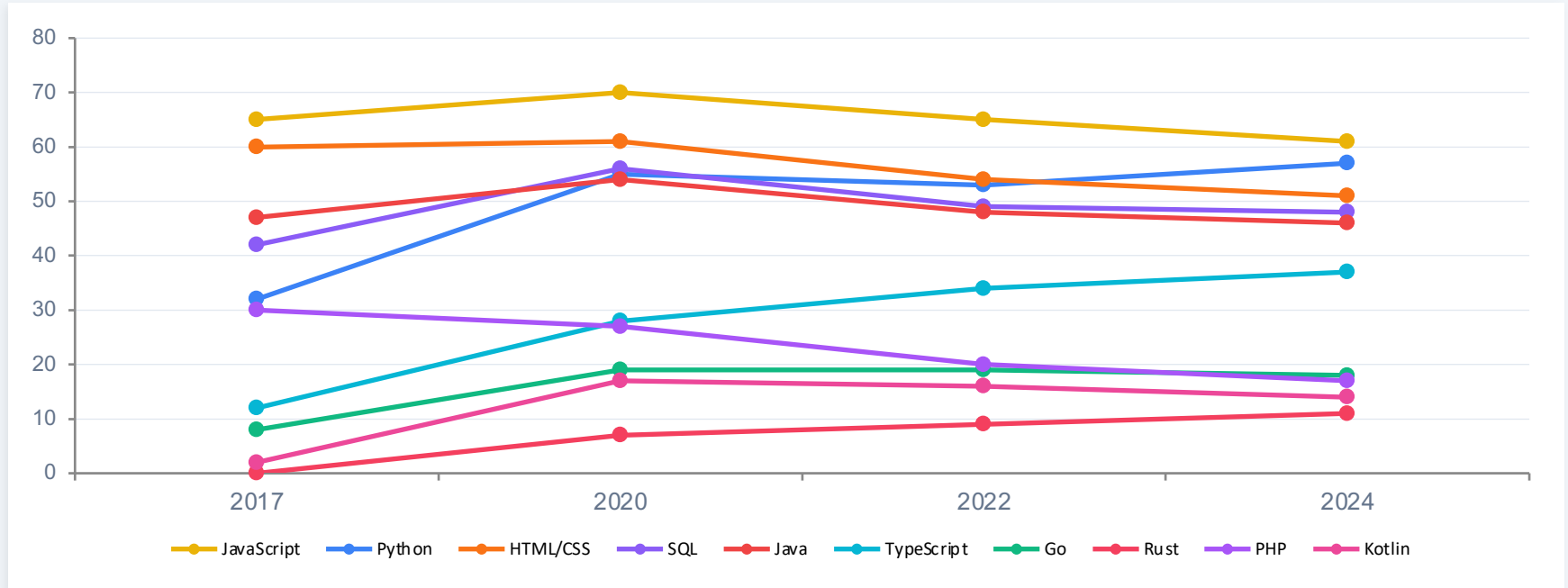
Based on JetBrains' annual survey of 23,262 developers worldwide

Covering programming languages, tooling preferences, AI sentiment, and career trends

Tracking 20 languages across 8 consecutive years (2017-2024)

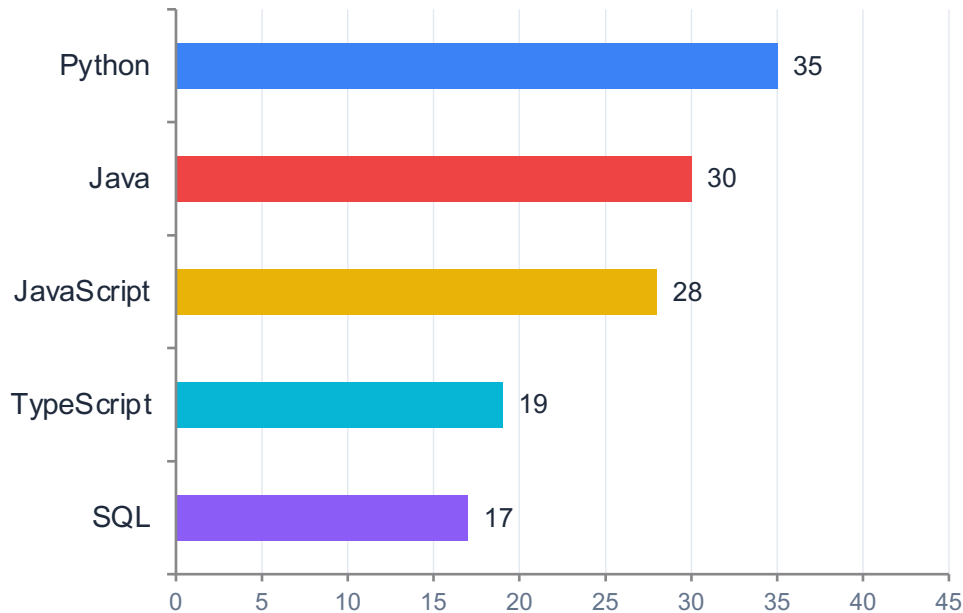
jetbrains.com/lp/devecosystem-2024

Python Closes the Gap: 8 Years of Language Usage Shifts



Key shifts: JS peaked 70% (2020) -> 61% • Python 32% -> 57% • TypeScript 12% -> 37% (3x) • PHP 30% -> 17%

Python Is Now the #1 Primary Language at 35%



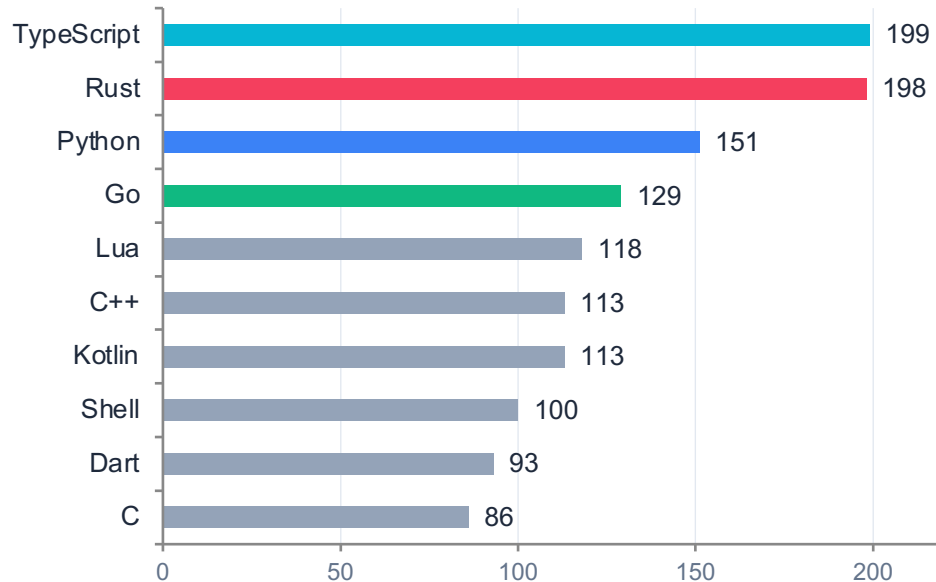
Adoption Intent

Developers rarely plan to switch their primary stack

● Python	6%
● TypeScript	5%
● Java	4%
● SQL	4%

Ecosystem lock-in keeps developers loyal to their chosen stacks

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



What Drives Satisfaction?

Type-safe, modern-tooling languages dominate the top spots

Type Safety

TypeScript & Rust combine compiler guarantees with expressive type systems

Modern DX

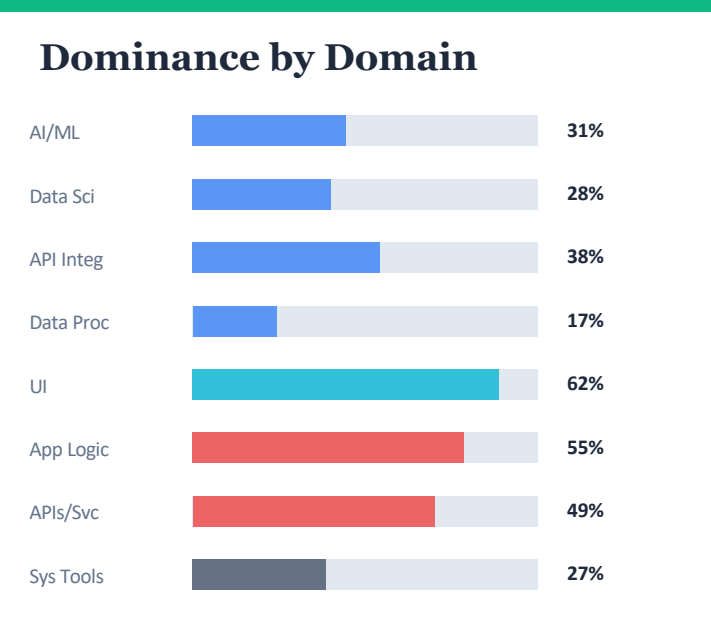
Strong tooling, package managers, and IDE support drive happiness

The Gap

Top 2 score 198+ vs. C at 86 - a 2.3x satisfaction spread

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Language	Share
AI / Machine Learning	Python	31%
Data Science & Analytics	Python	28%
Integrating with APIs	Python	38%
Data Processing	Python	17%
User Interface	TypeScript	62%
Application Logic	Java	55%
APIs & Services	Java	49%
System Tools & Components	C++	27%



Key insight: Language choice is driven by domain, not general preference. Python owns AI/data; TypeScript/JS own UI; Java owns backend; C++ owns systems.

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML teams

It is the undisputed leader - 31% AI/ML, 28% data science share

2

TypeScript is the future of frontend and full-stack

Highest satisfaction (199) and tripling usage since 2017

3

Rust and Go are the growth plays for infrastructure

Performance + safety guarantees for systems and cloud-native tooling

4

PHP and Ruby hiring pools are shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4% - plan succession

5

Satisfaction correlates with type safety and modern tooling

Factor this into retention strategy - top languages all offer strong DX